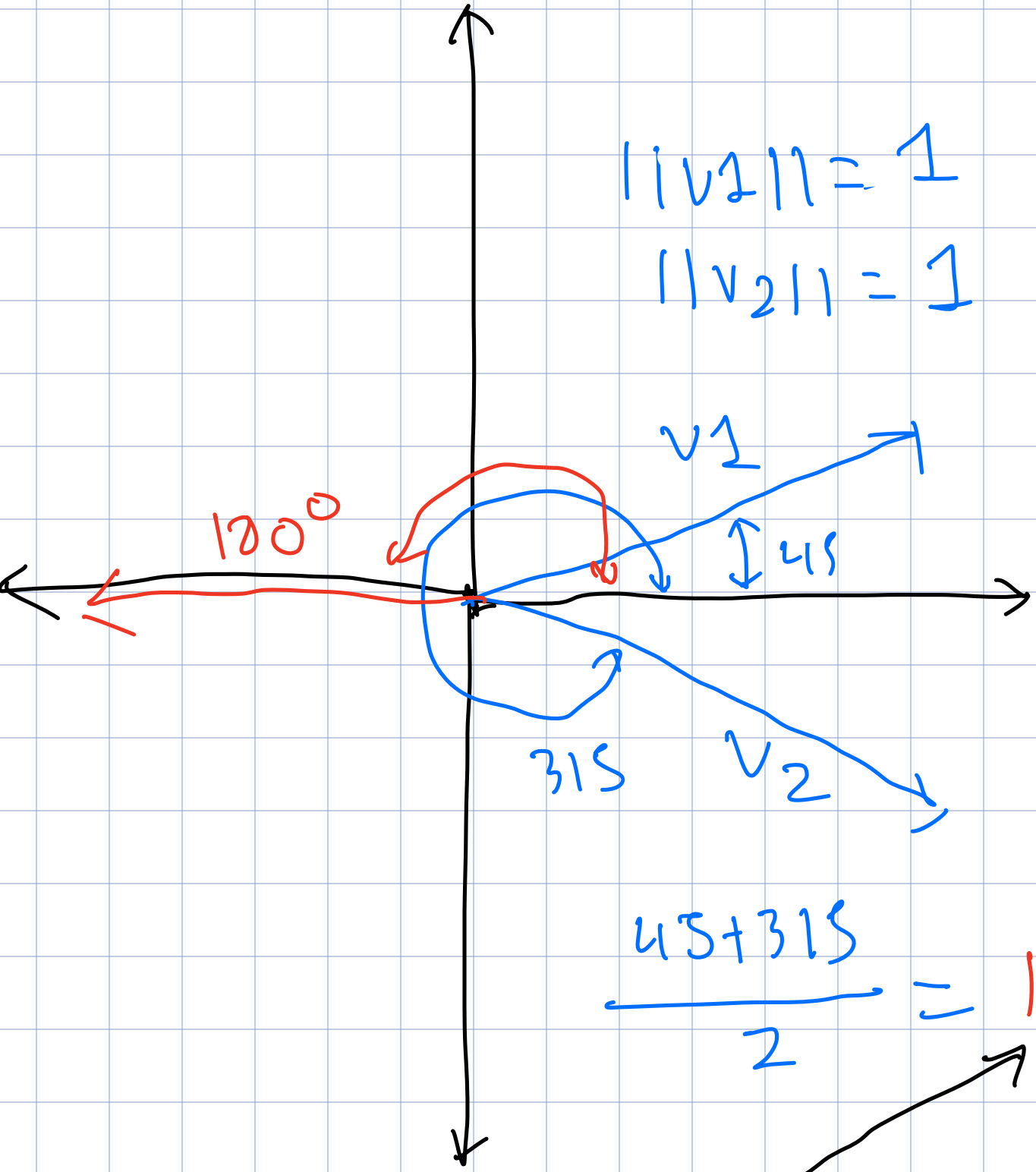
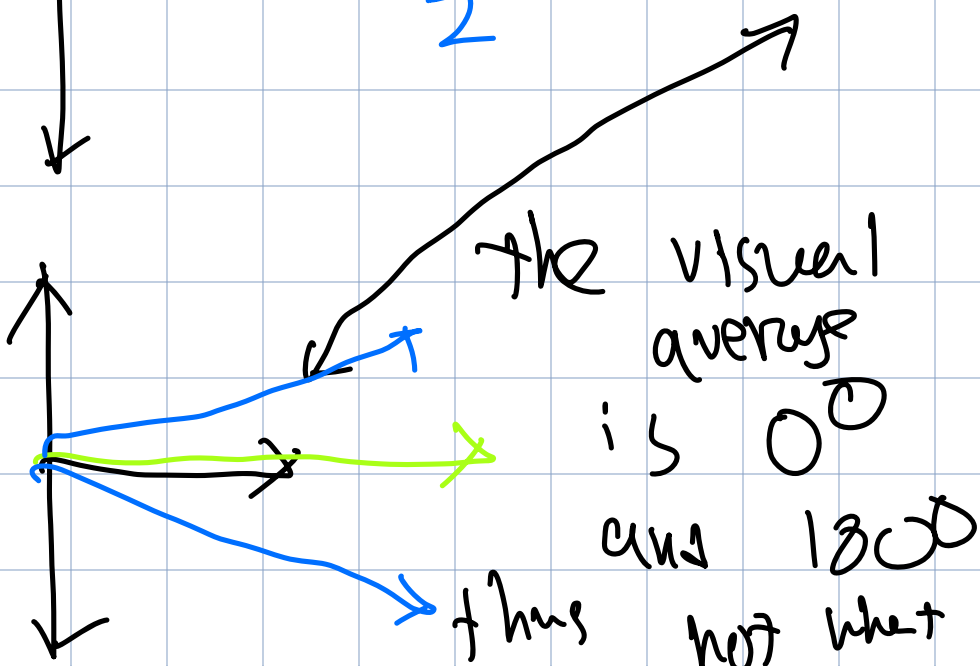
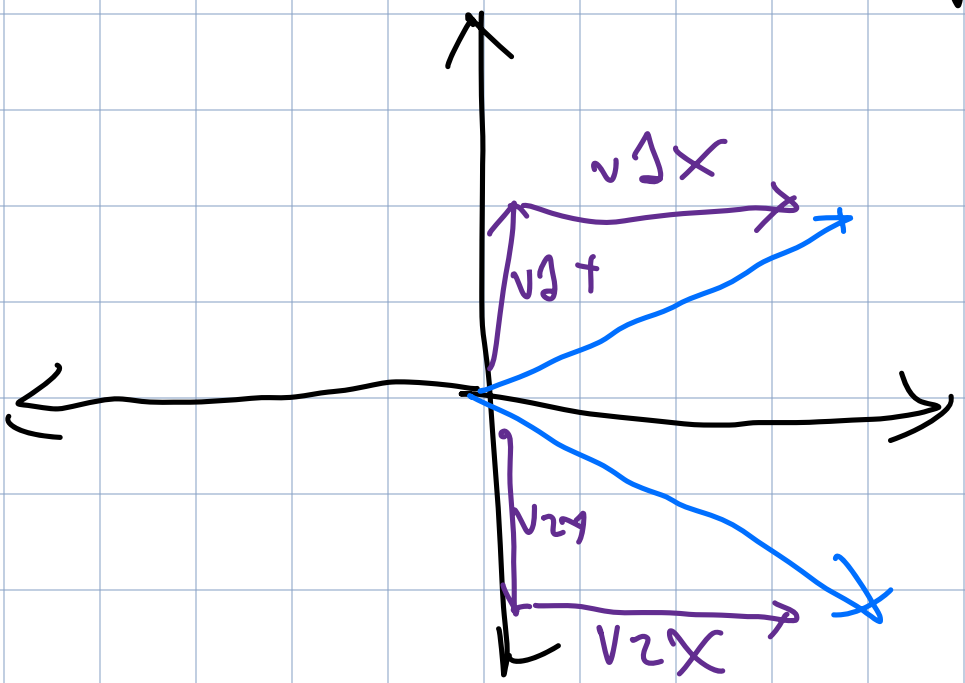


$$\frac{45+315}{2} = 180^\circ$$



we want

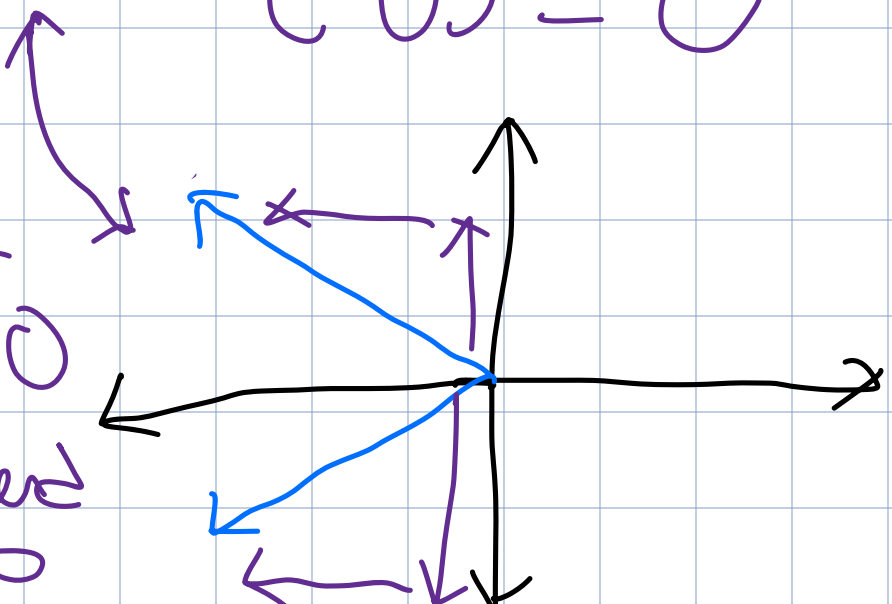


But if we sum the vectors by components and find the resultant vector's angle:

$$\arctan\left(\frac{v_{2y}+v_{1y}}{v_{2x}+v_{1x}}\right)$$

$$= \arctan(0) = 0$$

But arctan of this is 0 too instead of 180°



so we use atan2( )